Helix Forge Research Synthetic Sampling Methods Study

HELIX FORGE LAB / p04-full-004191 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded synthetic sampling evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000003 (research/literature-notes/2026-07-13/helix-forge-research/metric-contracts/tables/helix-forge-research-metric-contracts-metrics-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	232	0.874	0.856	0.865
slice-02	185	0.891	0.869	0.880
slice-03	228	0.873	0.882	0.877
slice-04	173	0.890	0.858	0.874
slice-05	211	0.872	0.871	0.871